



Oncor Skyview 345/138 kV Switch Project – ERCOT Independent Review Scope

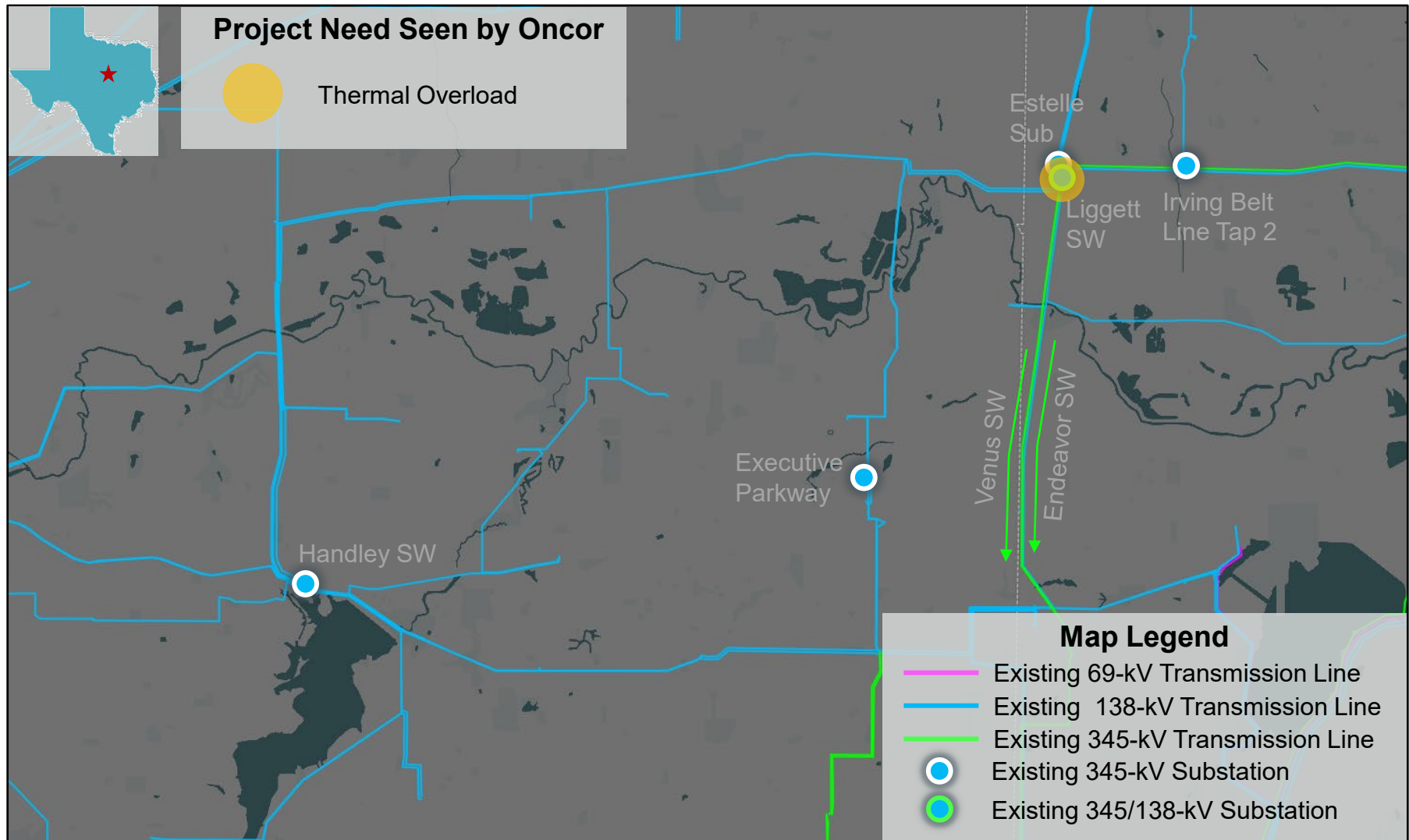
Tanzila Ahmed

RPG Meeting
October 28, 2025

Introduction

- Oncor submitted the Skyview 345/138 kV Switch Project for Regional Planning Group (RPG) review in September 2025
 - This Tier 2 project is estimated to cost approximately \$56.82 million and will require a Certificate of Convenience and Necessity (CCN) filing
 - Estimated in-service date (ISD) is May 2027
 - Addresses the thermal overloads and voltage violations due to proposed load additions in the Dallas and Tarrant Counties
- This project is currently under ERCOT Independent Review (EIR)

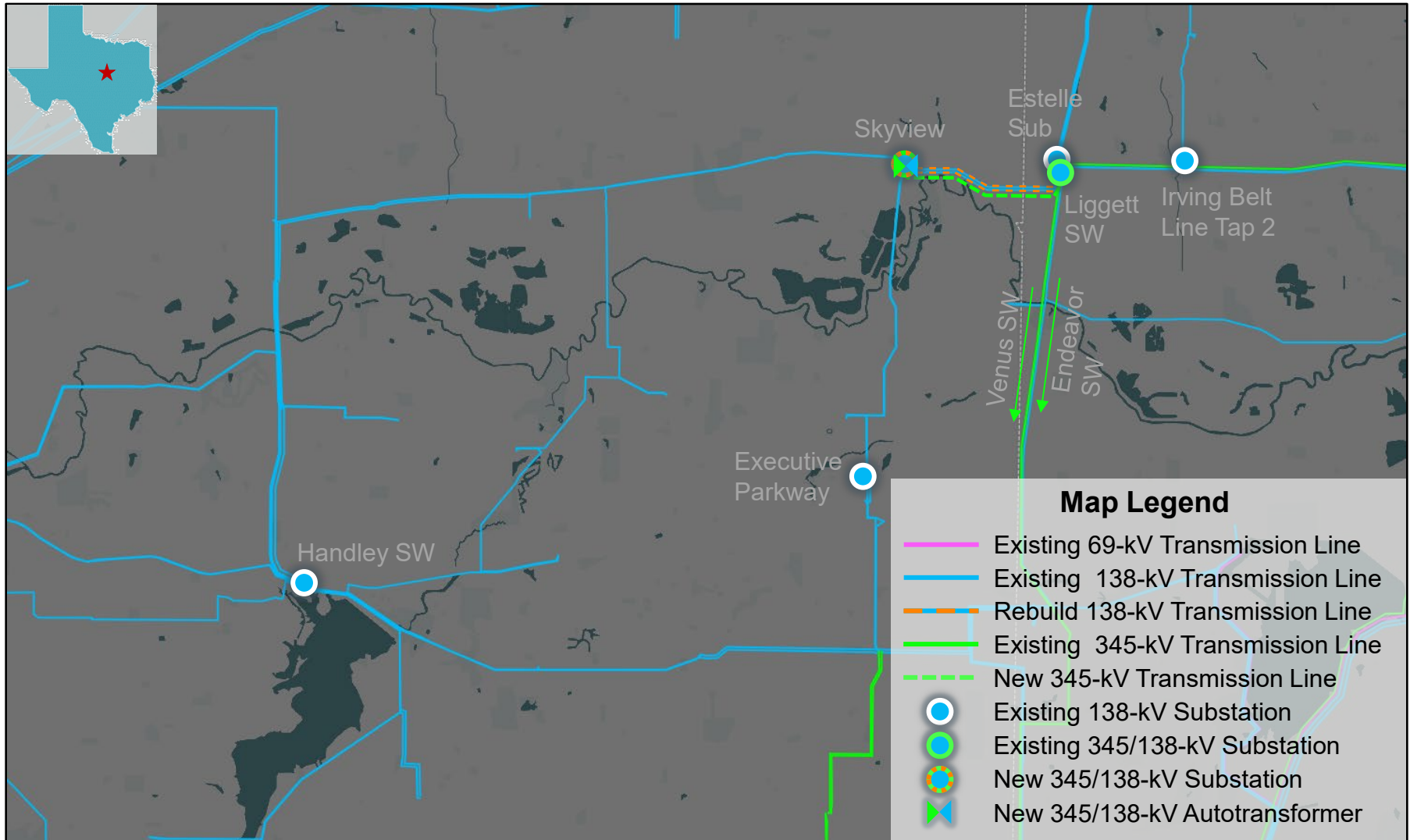
Study Area Map with Violations Seen by Oncor



Project Proposed by Oncor (cont.)

- Construct a new Skyview 345/138-kV Switch (SW) by installing one 345-kV, 3,200 A breaker and seven 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement
- Install one 600 MVA 345/138-kV Autotransformer at Skyview Switch with a normal and emergency ratings of 700 MVA and 750 MVA, respectively
- Rebuild portion of the existing Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line on triple-circuit capable with both circuit in place, with normal and emergency ratings of at least 614 MVA, approximately 2.6-mile
- Construct a new Liggett Switch to Skyview Switch 345-kV transmission line on the newly rebuild triple-circuit capable structures on the upgraded 2.6-mile portion of the exiting Liggett Switch to Handley Switch / Executive Parkway 138-kV double-circuit transmission line, with normal and emergency ratings of at least 1,912 MVA, approximately 2.6-mile;
- Install one 345-kV, 3,200 A circuit braker at Liggett 345-kV Switch
- For the 345-kV terminal equipment, ensure they meet or exceed a rating of 3,200 A
- For the 138-kV terminal equipment, ensure they meet or exceed a rating of 3,000 A

Map with Proposed Project by Oncor



Study Assumptions – Base Case

- Study Region
 - North Central Weather Zone, focusing on the transmission elements in Dallas and Tarrant Counties
- Steady-State Base Case
 - Final 2024 Regional Transmission Planning (RTP) 2027 summer peak, posted in Market Information System (MIS), will be updated to construct the summer peak load study base cases
 - Summer Peak Case: 2024RTP_2027_SUM_12202024
 - Link: <https://mis.ercot.com/secure/data-products/grid/regional-planning>

Study Assumptions – Transmission

- Based on the October 2025 Transmission Project and Information Tracking (TPIT) posted on ERCOT website, projects with in-service dates on or before May 2027 within the study area will be added to the study base case if not already modeled in the case
 - TPIT Link: <https://www.ercot.com/gridinfo/planning>
 - See [Appendix A1](#) for the list of transmission projects to be added
- Transmission projects identified in the 2024 RTP as placeholder projects related to this RPG project will be removed to develop the study base case
 - See [Appendix A2](#) for the list of transmission projects to be removed

Study Assumptions – Generation

- New generation that met Planning Guide Section 6.9(1) condition with Commercial Operation Date (COD) before May 2027 in the study area at the time of the study, but not already modeled in the RTP case, will be added to the study base case based on the September 2025 Generator Interconnection Status (GIS) report posted on the ERCOT website in October 2025
 - GIS Link: <https://www.ercot.com/gridinfo/resource>
 - See [Appendix B](#) for the list of generation projects to be added
- All generation will be dispatched consistent with the 2024 RTP methodology
- All recent retired/indefinitely mothballed units will be reviewed and opened (turned off), if not already reflected in the 2024 RTP Final case

Study Assumptions – Load & Reserve

- Load in study area
 - Loads in the study area will be maintained consistent with the 2024 RTP
- Reserve
 - Reserve will be maintained consistent with the 2024 RTP

Contingencies and Criteria

- Contingencies

- NERC TPL-001-5.1 and ERCOT Planning Criteria

Link: <https://www.ercot.com/mktrules/guides/planning/current>

- P0 (System Intact)
 - P1, P2-1, P7 (N-1 condition)
 - P2-2, P2-3, P4, and P5 (345-kV only)
 - P3: G-1+N-1 (G-1: generation outages) {Forney CCT, Handly SES, and Mt. Creek SES}
 - P6-2: X-1+N-1 (X-1: 345/138-kV transformers only) {Liggett Switch (SW), Endeavor SW, Venus SW, and Norwood SW}

- Criteria

- Monitor all 60-kV and above buses, transmission lines, and transformers in the study area (excluding generator step-up transformers)
 - Thermal
 - Use Rate A for pre-contingency conditions
 - Use Rate B for post-contingency conditions
 - Voltage
 - Voltages exceeding their pre-contingency and post-contingency limits
 - Voltage deviations exceeding 8% on non-radial load busses

Study Procedure

- Need Analysis
 - The reliability analysis will be performed to identify the need to serve the projected area load using the study base case
- Project Evaluation
 - Project alternatives will be tested to satisfy the NERC and ERCOT reliability requirements
 - ERCOT may also perform the following studies:
 - Planned maintenance outage
 - Long-Term Load-Serving Capability Assessment
 - Cost Estimate and Feasibility assessment will be requested from TSPs
- Additional Analysis on Preferred Option
 - Congestion analysis may be performed based on the recommended transmission upgrades to ensure that the identified transmission upgrades do not result in new congestion within the study area

Next Steps and Tentative Timeline

- Tentative Timeline
 - Status updates at future RPG meetings
 - Final recommendation – Q1 2026

Thank you!



Stakeholder comments also welcomed through:

Tanzila.Ahmed@ercot.com

Robert.Golen@ercot.com

Appendix A1 – Transmission Projects to Add

- List of transmission projects to be added to study base case

RPG/TPIT No	Project Name	Tier	Project ISD	From County
24RPG007	Canton Area Loop Project	Tier 2	10/1/2026	Van Zandt
24RPG016	Rand Area Loop Project	Tier 2	4/1/2027	Kaufman
24RPG017	Venus Switch to Sam Switch 345-kV Line Project	Tier 1	5/1/2026	Ellis, Hill
24RPG021	Forney 345/138-kV Switch Rebuild Project	Tier 1	12/1/2025	Kaufman
24RPG022	Wilmer 345/138-kV Switch Project	Tier 1	5/1/2026	Dallas

Appendix A2 – Transmission Projects to be Remove

- List of placeholder transmission projects that are directly related to proposed project were removed from the study base case

RTP Project ID	Project Name	County(s)
2024-NC77	Skyview (2231) 345/138-kV Substation Addition and Area Upgrade	Dallas, Tarrant

Appendix A2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

RTP Project ID	Project Name	County(s)
2024-NC06	Miller (824) to Newman (849) 138-kV Line Upgrade	Dallas
2024-NC07	Walnut (823) to Naaman (825) 138-kV Line Upgrade	Dallas
2024-NC08	Centerville (810) to Miller (824) 138-kV Line Upgrade	Dallas
2024-NC09	Walnut (823) to Newman (849) 138-kV Line Upgrade	Dallas
2024-NC10	Glen Heights (217) to Desoto Switch 8 (2424) 138-kV Line Upgrade	Ellis, Dallas
2024-NC11	Watermill (2427) to Tricorner (2432) to Trinidad (3123/3124) 345-kV Line Upgrades	Dallas, Henderson
2024-NC14	Green Road (3069) to Ten Mile (2126) to Watermill (2429) to Reindeer (3065) 138-kV Line Upgrades	Dallas
2024-NC16	Loop 9 (2847) to Watermill (2427) 345-kV Line Upgrade	Dallas
2024-NC21	Carrollton Northwest (2363) to South TNP (37100) to TI TNP (37080) 138-kV Line Upgrades	Dallas, Denton
2024-NC41	Ben Davis (968) to Gob Lane (2497) 138-kV Line Upgrade	Dallas
2024-NC63	Watermill Area 345-kV Line Additions and Reactive Support	Dallas, Navarro, Hill, Leon
2024-NC67	Elkton (3105) to Tri Corner (2432) 345-kV Line Additions and Upgrades	Dallas, Kaufman, Smith, Van Zandt

Appendix A2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

RTP Project ID	Project Name	County(s)
2024-NC68	Batchler Road (2217) to Watermill (2427) 345-kV Line Upgrades	Dallas
2024-NC70	Miller Road (2632) 345/138-kV Substation Addition and 345-kV Lines Retermination	Dallas, Ellis
2024-NC71	Mitchell Bend Switch (1895) to Carmichael Bend Switch (2285) to Benbrook Switch (1873) 345-kV Line Upgrades	Tarrant, Hood, Grayson
2024-NC73	Greene Road (3069) to Cedar Crest (2486) 138-kV Line Upgrades	Dallas
2024-NC75	Primrose Fort Worth (2196) to Rocky Creek (1881) to Primrose Fort Worth (2197) 138-kV Line Upgrades	Tarrant
2024-NC79	Benbrook Switch 345/138-kV Transformer (1869/18741 and 1870/18751) Upgrades	Tarrant
2024-NC80	Desoto Switch 345/138-kV Transformer Addition (2431/2424) and Desoto Switch (2424) to Loop9 (2848) 138-kV Line Addition	Dallas
2024-NC82	Rocky Creek (1880/1881) 345/138-kV Transformer Addition	Tarrant
2024-NC83	Hempill (2164) to Mistletoe Heights (2173) 138-kV Line Upgrade	Tarrant
2024-NC84	Wedgewood Switch (2184) to Bryant Irvin (2182) 138-kV Line Upgrade	Tarrant

Appendix B – New Generation Projects to Add

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
21INR0359	Hickerson Solar	SOL	1/30/2025	311.1	Bosque
22INR0437	Tormes Solar	SOL	2/13/2025	356.6	Navarro
23INR0266	Langer Storage	OTH	7/15/2025	61.53	Bosque
23INR0538	Roadrunner Crossing BESS SLF	OTH	4/28/2025	150.4	Eastland
24INR0181	Bynum Solar Project	SOL	4/28/2025	56.0	Coryell
24INR0188	Tehuacana Creek Solar	SOL	2/12/2025	505.4	Navarro
24INR0189	Tehuacana Creek BESS SLF	OTH	2/12/2025	419.0	Navarro
24INR0364	Pitts Dudik II	SOL	2/12/2025	30.2	Hill
24INR0631	Radian Storage SLF	OTH	7/24/2024	160.0	Brown
25INR0018	Yellow Cat Wind	WIN	2/12/2025	262.0	Navarro
25INR0101	Mallard BESS	OTH	7/16/2025	257.3	Collin
25INR0166	Padrino Solar	SOL	3/4/2025	201.4	Hill
25INR0199	Bonham Solar 1	SOL	5/9/2025	138.4	Limestone
25INR0229	OCI Cobb Creek Solar	SOL	6/11/2025	203.1	Hill
25INR0231	Apache Hill BESS	OTH	7/17/2024	201.2	Hood
25INR0233	OCI Cobb Creek ESS	OTH	6/11/2025	201.6	Hill
25INR0391	Purple Sage BESS 1	OTH	2/13/2025	156.0	Collin
25INR0392	Purple Sage BESS 2	OTH	2/13/2025	156.0	Collin
26INR0543	Three Canes Solar SLF	SOL	2/12/2025	333.0	Navarro